King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $f(x) = \dfrac{x^3 - 3x^2 - 4}{x^2}$ then $f'(1) =$

(a) 9

(b) 6

(c) -5

(d) -7

(e) 5

2. If $f(x) = \begin{cases} x^2 + 3 & x \neq 1 \\ 2 & x = 1 \end{cases}$ then $f(1) - \lim_{x \to 1} f(x) =$

(a) 0

(b) does not exist

(c) 2

(d) -2

(e) 3

3. Let $f(x) = \begin{cases} x^3 & x \leq 2 \\ ax^2 & x > 2 \end{cases}$ The value of a that makes f continuous is

(a) -2

(b) 2

(c) 3

(d) -3

(e) 1

4. The equation of the tangent line to the curve $y = \sqrt[3]{x-1} + 2$ at the point $(1, 2)$ is

(a) $x = 1$

(b) $y = 1$

(c) $y = x$

(d) $y = 2$

(e) $x = 2$

5. $\lim_{x\to-\infty}\left(\frac{2}{x+1}-\frac{x+1}{2}\right)=$

(a) -4

(b) $-\infty$

(c) ∞

(d) 4

(e) 0

6. $\lim_{x\to 64}\frac{16-x^{1/3}}{16-\sqrt{x}}=$

(a) $\dfrac{1}{2}$

(b) $\dfrac{3}{2}$

(c) 0

(d) $-\infty$

(e) does not exist

7. $\lim_{x\to 0^+}\left(\frac{1}{x}-\sin\frac{1}{x}\right)=$

(a) 1

(b) -1

(c) $-\infty$

(d) $+\infty$

(e) 0

8. $\lim_{x\to 2^-}\frac{|x-2|}{x-2}+\lim_{x\to 2^+}\frac{x-2}{|x-2|}=$

(a) 1

(b) does not exist

(c) -2

(d) 2

(e) 0

9. $\lim_{x\to 0} \frac{x-\cos x+1}{2x+\sin x} =$

(a) $-\frac{1}{2}$

(b) $\frac{2}{3}$

(c) does not exist

(d) 1

(e) $\frac{1}{3}$

10. Let $f(x) = \begin{cases} \frac{1}{2}x+1, & x \le 2 \\ 3-x, & x > 2 \end{cases}$ then

(a) $\lim_{x\to 2} f(x)$ exists.

(b) f is continuous at $x = 2$.

(c) $\lim_{x\to 2^+} f(x) = f(2)$.

(d) f has a non-removable discontinuity at $x = 2$.

(e) f has a removable discontinuity at $x = 2$.

11. $\lim_{x\to -\infty} \frac{\sqrt{3x^6+\cos x}}{2x^3+\sin x} =$

(a) $-\infty$

(b) $+\infty$

(c) $-\sqrt{3}$

(d) $-\frac{\sqrt{3}}{2}$

(e) $-\frac{\sqrt{3}}{3}$

12. The number of vertical asymptotes the function $f(x) = \frac{x}{\tan x}$ has in the interval $[-\pi, \pi]$ is

(a) 3

(b) 2

(c) 4

(d) 0

(e) 1

13. If the points $P_1 = (1, 1)$, $P_2 = (3, y_1)$ and $P_3 = (5, y_2)$ lie on the graph of $y = f(x)$ and the slope of the secant line through P_1 and P_2 is 2 and the slope of the secant line through P_1 and P_3 is -1, then $f(3) + f(5) =$

(a) 4

(b) -2

(c) 2

(d) -1

(e) 1

14. The slope of the tangent line to the curve $y = \dfrac{1}{x-1}$ at the point $\left(3, \dfrac{1}{2}\right)$ is

(a) $\dfrac{1}{4}$

(b) $\dfrac{1}{3}$

(c) $\dfrac{1}{9}$

(d) $-\dfrac{1}{9}$

(e) $-\dfrac{1}{4}$

15. $\lim_{x\to -2} \frac{x^3+8}{x^4-16} =$

(a) does not exist

(b) 0

(c) $\dfrac{1}{2}$

(d) $-\dfrac{1}{2}$

(e) $-\dfrac{3}{8}$

16. Which one of the following statements is always true?

(a) $\lim_{x\to 0} \frac{\cos x - 1}{x} = -1$.

(b) If $\lim_{x\to a^-} f(x) = -\infty$, then $x = a$ is a vertical asymptote.

(c) If $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ do not exist, then $\lim_{x\to a} f(x)g(x)$ does not exist.

(d) $f(x) = \left\lfloor \dfrac{1}{2} - x \right\rfloor$ is continuous on $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right)$.

(e) If $\lim_{x\to a^+} f(x) = f(a) = \lim_{x\to a^-} f(x)$, then f is differentiable at a.

17. If $y = x + 4$ is the tangent line to $f(x) = k\sqrt{x}$ at $(a, f(a))$, then $a + k =$

(a) 6

(b) 8

(c) 2

(d) 4

(e) 15

18. The sum of all values of x in $[0, 2\pi)$ at which the function $f(x) = x + \cos x$ has a horizontal tangent line is equal to

(a) π

(b) 3π

(c) $\dfrac{3\pi}{2}$

(d) 2π

(e) $\dfrac{\pi}{2}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$f = x - 3 - 4x^{-2}$, $f' = 1 + 8x^{-3}$, $f'(1) = 9$.
2	**(d)**	$f(1) = 2$, limit $= 1^2 + 3 = 4$, difference -2.
3	**(b)**	$2^3 = a \cdot 2^2$, so $a = 2$.
4	**(a)**	Difference quotient $= (x-1)^{-2/3} \to \infty$: vertical tangent $x = 1$.
5	**(c)**	First term $\to 0$, $-\frac{x+1}{2} \to +\infty$, so limit ∞.
6	**(b)**	No indeterminacy: $\frac{16-4}{16-8} = \frac{3}{2}$.
7	**(d)**	$\frac{1}{x} - \sin\frac{1}{x} \geq \frac{1}{x} - 1 \to \infty$.
8	**(e)**	First limit -1, second $+1$, sum 0.
9	**(e)**	Divide by x: $\frac{1+0}{2+1} = \frac{1}{3}$.
10	**(d)**	Left limit 2, right limit 1: jump, non-removable.
11	**(d)**	$\sqrt{x^6} = -x^3$ for $x < 0$, so limit $= -\frac{\sqrt{3}}{2}$.
12	**(b)**	Asymptotes only at $x = \pm\pi$; none at 0 or $\pm\frac{\pi}{2}$: 2.
13	**(c)**	$f(3) = 1 + 2 \cdot 2 = 5$, $f(5) = 1 - 4 = -3$, sum 2.
14	**(e)**	$f'(x) = -\frac{1}{(x-1)^2}$, so $f'(3) = -\frac{1}{4}$.
15	**(e)**	Cancel $x + 2$: $\frac{x^2-2x+4}{(x-2)(x^2+4)} \to \frac{12}{-32} = -\frac{3}{8}$.
16	**(b)**	Only the vertical-asymptote statement is always true; others have counterexamples.
17	**(b)**	$k = 2\sqrt{a}$, $2a = a + 4$, $a = 4$, $k = 4$, so $a + k = 8$.
18	**(e)**	$f' = 1 - \sin x = 0 \Rightarrow x = \frac{\pi}{2}$ only.